



Feature

How successful are AI-discovered drugs in clinical trials? A first analysis and emerging lessons

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AI techniques are making inroads into the field of drug discovery. As a result, a growing number of drugs and vaccines have been discovered using AI. However, questions remain about the success of these molecules in clinical trials. To address these questions, we conducted a first analysis of the clinical pipelines of AI-native Biotech companies. In Phase I we find AI-discovered molecules have an 80–90% success rate, substantially higher than historic industry averages. This suggests, we argue, that AI is highly capable of designing or identifying molecules with drug-like properties. In Phase II the success rate is ~40%, albeit on a limited sample size, comparable to historic industry averages. Our findings highlight early signs of the clinical potential of AI-discovered molecules.

Keywords: AI drug discovery; AI-native Biotech; AI-discovered drug; Clinical trial success rate; AI drug in clinical trials

Introduction

Drug discovery is a lengthy and costly process, with a high degree of uncertainty. Despite latest-generation experimental technologies, many discovery programs struggle. Even if successful, programs often take many years to complete. This is particularly true for new small molecules, which typically take 4–6 years to discover.^(p1) Other modalities, such as biologics and vaccines, can be somewhat faster and more successful but discovery remains complex and risky.^(p2)

AI promises to revolutionize drug discovery,^(p3) by conducting many of the most time-consuming, repetitive and costly steps *in silico*, and by vastly increasing the scale of exploration. Prominent examples of AI techniques used in drug discovery include: knowledge graphs to mine OMICs and other data to understand disease biology and identify drug targets and biomarkers^(p4); the use of generative AI to design small molecules^(p5); optimising the design of antibodies and other proteins using AI-powered structure

prediction algorithms, such as AlphaFold^(p6); AI-powered repurposing of drug molecules^(p7); and many others.^{(p3),(p8),(p9)}

Over the past decade, following the introduction of these techniques, the number of AI-discovered drug and vaccines molecules has increased substantially. In 2022, we showed that AI-discovered small-molecule numbers were growing exponentially and beginning to match the number of classically discovered small molecules.^(p10) This trend has continued

since then.^(p11) Similar exponential growth can be seen in AI-discovered biologics, although the number of molecules is still smaller.^(p12) We also see signs that AI is beginning to accelerate drug discovery timelines.^(p10)

The pharmaceutical industry has embraced the use of AI in R&D.^(p13) By early 2024, each of the top 20 pharmaceutical companies had announced activities in the space. A substantial proportion of these activities take place as collaborations between pharmaceutical companies and biotechnology companies specialising in AI (the so-called AI-native Biotechs). As a result, partnership deals in the AI space have increased enormously in number and size over the past 5 years.^(p13) Despite the strong progress in the field, many open questions remain. Perhaps most important are questions about the quality of AI-discovered molecules, especially their safety and efficacy in clinical trials.^(p14) To begin answering some of these questions, we have conducted a first analysis of the industry-wide pipeline of AI-discovered drugs and vaccines, focusing specifically on clinical success rates. Because of the limited number of these molecules in clinical trials and the rapid evolution of the field, this is very much a preliminary analysis which, over time, will need to be confirmed. However, given the importance of the topic to the industry, we believe it is crucial to report this early evidence and discuss its potential implications.

Analysis of AI-discovered molecules in clinical trials

To understand AI-discovered molecules in clinical trials, we reviewed the pipelines of AI-native Biotech companies using publicly available databases (see [supplementary material](#) methods and information online). Focusing on AI-natives, we believe, is an appropriate proxy for the industry overall, because a substantial proportion of the AI-powered drug discovery work takes place in these companies.^{(p11),(p13)} Also, because many of the clinical-stage AI-native Biotechs have partnerships with larger pharma companies, our approach provides a representative view across the industry. A full description of the methodology is given in the supplementary

methods. Briefly, starting from a compendium of >100 AI-native Biotech companies (see [supplementary material Table S1](#) online), we gathered pipeline data from publicly available sources. These data were crosschecked and, where appropriate, refreshed based on the latest published readouts of clinical trials (see [supplementary material Table S2](#) online).

Each AI-native Biotech was categorized based on the primary focus of their AI techniques, as described in publications and conference presentations. We focused this categorization on the mode-of-discovery (i.e., how AI was used to discover or identify the respective molecules). Specifically, we distinguished the following: (i) molecules with AI-discovered drug targets; (ii) AI-discovered small molecules; (iii) AI-discovered biologics; (iv) AI-discovered vaccines; and (v) AI-repurposed molecules. We have chosen this categorization based on the hypothesis that clinical success might differ by mode-of-discovery.

Our analysis shows that, since 2015, AI-native Biotechs and their partners in the pharmaceutical industry have entered 75 molecules into the clinic, of which 67 were in ongoing trials as of 2023. Over the past 10 years this number has grown exponentially, with year-over-year compound growth in excess of 60% ([Figure 1a](#)). This suggests that the ‘coming wave’ of AI in R&D, which is already well-documented in drug discovery,^{(p10),(p11)} is now occurring at the clinical trial stage. Unsurprisingly, most of the AI-discovered molecules are currently in Phase I trials, although some have already progressed to Phase II and beyond. These molecules represent a broad range of therapeutic areas with oncology particularly prominent, accounting for ~50% of AI-discovered molecules in Phase I and Phase II.

We also observe an increasingly diverse range of modes-of-discovery. Before 2020, AI-repurposed molecules dominated but have levelled off since then, constituting ~15% of the cross-industry clinical pipeline in 2023 ([Figure 1b](#)). By contrast, other modes-of-discovery have accelerated. At present, AI-discovered small molecules have the largest representation (>30% in 2023), with vaccines (~10%) and antibodies (~5%) representing a smaller share.

Lastly, some molecules with AI-discovered targets have entered the clinic, representing >30% of the pipeline in 2023. We note that many of the latter are also small molecules.

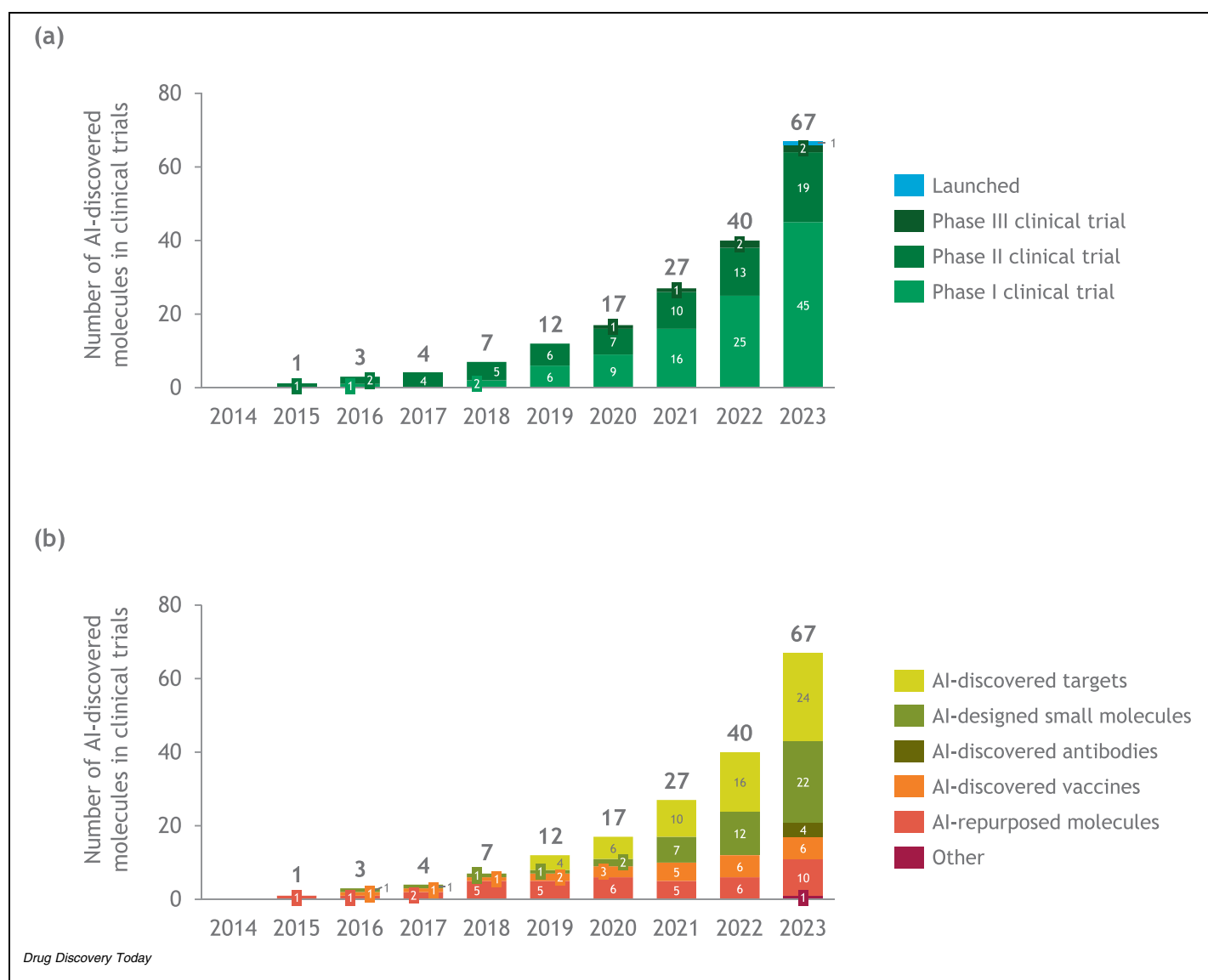
Following on, we conducted a preliminary analysis of clinical trial success rates of AI-discovered molecules. As of 2023 December, 24 AI-discovered molecules had completed Phase I trials, of which 21 were successful ([Figure 2a](#)). This suggests a success rate of 80–90%, which is substantially higher than historical industry averages that range from ~40%^(p15) to ~55–65%.^{(p1),(p16),(p17)} When we break out the Phase I data by mode-of-discovery we see similar results across the board ([Figure 2b](#)).

In Phase II, ten AI-discovered molecules have completed trials of which four were successful. This implies a success rate of 40%, which is in line with historical industry averages of 30–40%.^{(p1),(p15),(p16),(p17)}

It is important to note three caveats of this analysis. (i) The sample size is small, especially for the clinical success rates. As more data become available in the coming years, these numbers are likely to change substantially. In the meantime, readers might find it helpful to look at case studies of specific AI-discovered molecules, for example in recent reports.^(p11) (ii) The analysis includes only molecules from AI-native Biotechs, including those partnered with larger pharma companies, but it does not include AI-discovered molecules that originate purely from within larger pharma companies. Our data are therefore not fully exhaustive but, as discussed above, we believe they could be representative. (iii) The categorization of molecules is not mutually exclusive. Some of the molecules were discovered using multiple AI techniques but we could allocate them to only one category. We have called out these nuances in the discussion, insofar as they affect the conclusions we draw.

Implications for AI-powered drug discovery

Our analysis suggests that, in Phase I trials, AI-derived molecules can have a success rate of 80–90%, which is substantially higher success rates than historic averages.^{(p1),(p15),(p16),(p17)} This improvement in success rates could be due to several different reasons. One reason could be that AI

**FIGURE 1**

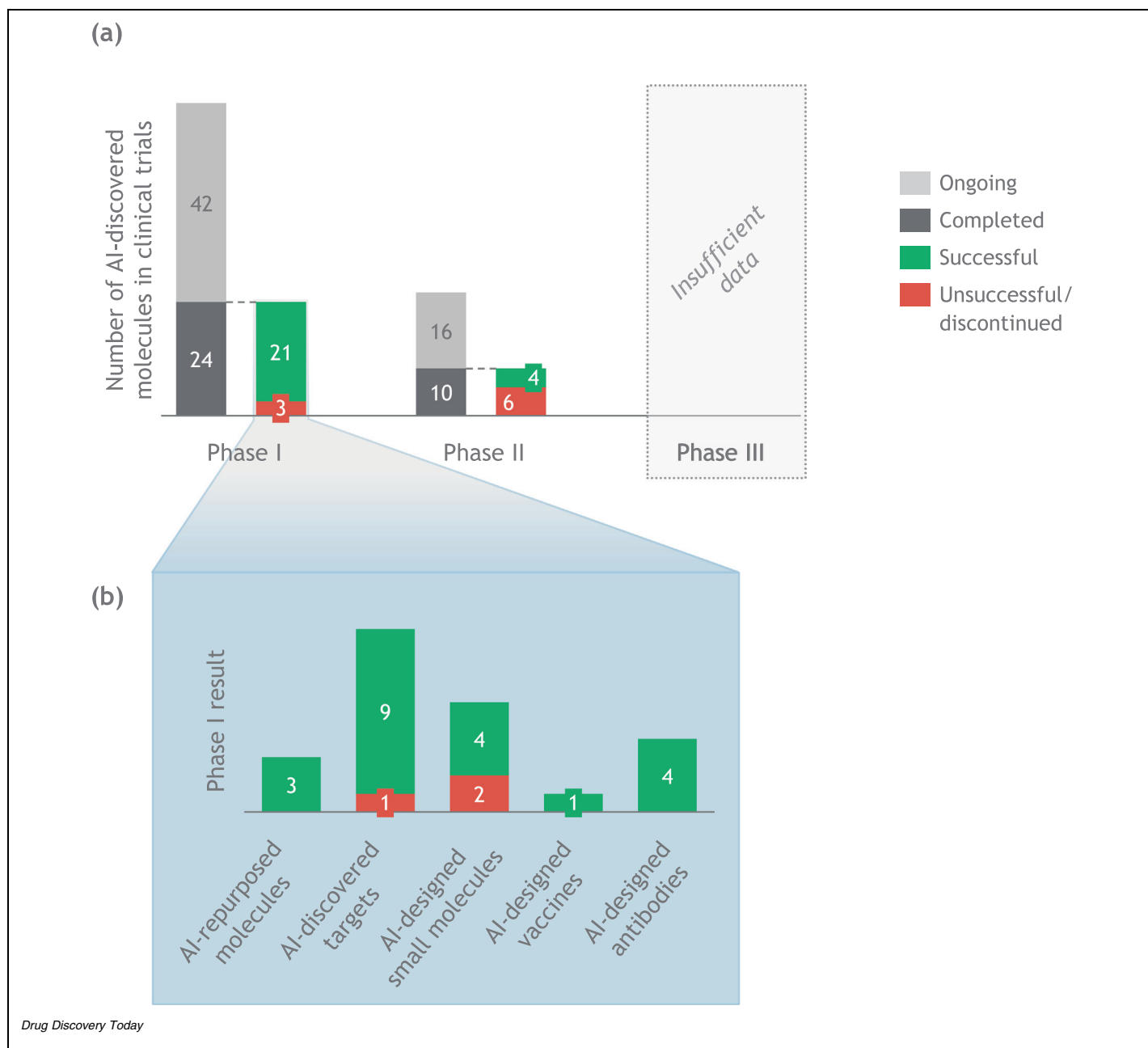
Number of molecules discovered by AI-first Biotechs that have entered clinical trials. The analysis includes molecules that were partnered with pharmaceutical companies and excludes COVID-19-related molecules. **(a)** AI-discovered molecules by clinical Phase. **(b)** AI-discovered molecules by mode-of-discovery.

discovery efforts pursue well-validated biological targets and pathways, which reduces the risk of on-target toxicity. Although this might play a part, we are seeing early signs of molecules going after novel targets passing through Phase I (we see at least three such molecules in our analysis which were all successful in Phase I, plus possibly more for which targets are not yet disclosed; see [supplementary material Table S3](#) online). Therefore, we believe that the high Phase I success rate is not just the result of going after well-validated biology.

Alternatively, the high Phase I success rate could be explained by the AI algorithms having been trained on data from well-established molecule series, which the algorithms fine-tune and optimize very effectively. At present we do not have sufficient information to assess this hypothesis but earlier analysis suggests that AI algorithms have been used to explore novel chemical space^(p10) and not just to fine-tune previously known structures.

Finally, the high Phase I success rate could be explained by the AI algorithms

in general being very capable of designing or selecting drug-like molecules, including novel molecules with optimized ADME and safety profiles, leading to fewer Phase I dropouts. Given the high success rate across different modes-of-discovery ([Figure 2b](#)), we believe this is likely to be at least part of the explanation. Although it is too early to claim that AI algorithms have solved the problem of molecule design in a general sense, the findings we report could provide a preview of what is possible. It is also interesting to note that, when looking at the three AI-discovered

**FIGURE 2**

The success of AI-discovered molecules in clinical trials so far. The analysis includes molecules that were partnered with pharmaceutical companies and excludes COVID-19-related molecules. **(a)** Clinical success of AI-discovered molecules by clinical Phase. **(b)** AI-discovered molecules that have completed Phase I trial, by mode-of-discovery.

molecules that failed in Phase I, only one was due to not achieving the evaluation criteria. The other two were discontinued owing to business decisions and pipeline reprioritisation (see [supplementary material Table S2](#) online).

In Phase II trials, our data indicate a success rate of AI-discovered molecules of ~40%, which is in line with historic indus-

try averages.^{(p1),(p15),(p16),(p17)} Because Phase II typically involves the proof of a biological or mechanistic concept, this might suggest that AI algorithms can identify disease-relevant targets and pathways but there is scope for improvement. Also, ‘the hard problem of drug discovery’,^(p18) namely achieving clinical efficacy, remains challenging. However, when looking at

the Phase II data of AI-discovered molecules more closely, we see a more nuanced picture: of the six candidates that were stopped or discontinued after Phase II only two were the result of negative outcome data – the other four candidates were stopped owing to shifts in business priorities, clinical operations challenges or other reasons. We also note that, for Phase II dis-

continuation due to business reasons, the current economic and funding challenges of Biotech companies could be an exacerbating factor. In particular, the tough business environment over the past few years^(p19) and regulatory changes, such as the Inflation Reduction Act in the USA,^(p20) might have led to portfolio reprioritisation decisions that were unrelated to the underlying AI techniques.

Concluding remarks and future outlook

Our analysis gives a first view of the potential of AI-discovered molecules in clinic trials, and perhaps provides a glimpse into the future of AI-powered R&D. To explore the industry-wide implications of these findings, let us conduct a thought experiment. If we take, at face value, the success rates we observed for AI-discovered molecules in Phase I and II and assume that these hold in the future, and if we combine these with historic Phase III success rates, a striking picture emerges: the probability of a molecule succeeding across all clinical phases end-to-end would increase from 5–10%^{(p1),(p15),(p16),(p17)} to ~9–18%. This would represent almost a doubling of pharmaceutical R&D productivity overall, which would bring enormous benefits. It would enable companies either to achieve the same output with fewer resources and cost, or to increase the total number of new drugs launched within the same resources.

Beyond what we can already observe in our analysis, there is reason to believe that AI techniques could further improve clinical performance, especially in Phase II and III trials. Understanding the drivers of disease and identifying and validating drug targets is an area where many AI-native Biotechs, pharmaceutical companies and academic institutions are actively investing. Efforts are particularly focused on OMICs and phenotypic data generation,^(p21) reverse translation,^(p22) novel patient-derived models^(p23) and applications of large language models to better mine disease data.^(p24) These technologies can help bridge the gap between molecule design and clinical efficacy, and further improve clinical trial success rates beyond historic averages. Ultimately, the promise of AI in drug discovery is to bring more-innovative medicines to patients faster, better and cheaper. We have already started seeing the speed and cost impacts

in preclinical workflows from these techniques.^{(p10),(p25)} Our findings show that benefits are beginning to manifest in clinical trials as well. Despite the statistical caveats, we believe these results paint an intriguing picture for the future of R&D. In the coming years, as more clinical results for AI-discovered molecules become available, it will be exciting to see how AI techniques will impact R&D productivity overall.

Competing interests

The authors of this article are employees of Boston Consulting Group (BCG), a management consultancy that works with the world's leading biopharmaceutical companies. The research for this specific article was funded by BCG's Health Care practice.

CRedit authorship contribution statement

Madura KP Jayatunga: Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis, Conceptualization. **Margaret Ayers:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Lotte Bruens:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Dhruv Jayanth:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Christoph Meier:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Conceptualization.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.drudis.2024.104009>.

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